The Exponential Formula and Three Functional Identities of A092635

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Abstract

We prove the Euler-polynomial exponential formula proposed for A092635 and all three functional identities conjectured in its 2024 update. The arguments use the formal asymptotic expansion of the gamma ratio defining the sequence, oddness of its logarithm, and the gamma recurrence. The identities hold to every order; no convergence of the generally divergent generating series is assumed.

1 Definition and exponential formula

OEIS A092635 [1] defines the coefficients a_j by

$$F(t) = \frac{\pi t}{16^t}\binom{2t}{t}^2 \sim G\big(1/(8t)\big), \qquad G(x) = \sum_{j\ge 0} a_j x^j, \quad a_0 = 1.$$

All series identities below are formal. Define the integers b_j by

$$B(z) = \frac{2}{1+e^{4z}} = \sum_{j\ge 0} b_j \frac{z^j}{j!}.$$

The entry proposes $G(x) = \exp(\sum_{j\ge 1} b_j x^j/j)$ and its coefficient recurrence; it separately conjectures three identities for $H(x) = xG(x)$.

Theorem 1. *In $\mathbb{Q}[[x]]$,*

$$G(x) = \exp\left(\sum_{j\ge 1} \frac{b_j}{j} x^j\right), \qquad a_m = \frac{1}{m}\sum_{j=1}^{m} b_j a_{m-j} \quad (m \ge 1).$$

Here $b_j = (-4)^j E_j(1)$, with Euler polynomials defined by $2e^{uz}/(e^z+1) = \sum E_j(u) z^j/j!$.

Proof. Let B_{2k} be the Bernoulli numbers in Stirling's logarithmic expansion [2],

$$\log\Gamma(t+1) \sim (t+\tfrac12)\log t - t + \tfrac12\log(2\pi) + \sum_{k\ge 1} \frac{B_{2k}}{2k(2k-1)t^{2k-1}}.$$

Substitution in $\log F(t)$ cancels the elementary terms and gives

$$\log G(x) = \sum_{k\ge 1} \frac{2B_{2k}(2^{-(2k-1)}-2)8^{2k-1}}{2k(2k-1)} x^{2k-1}.$$

On the other hand, use $2/(1+e^w) = 2/(e^w-1) - 4/(e^{2w}-1)$ and the Bernoulli generating function. For $j \ge 1$ it gives

$$b_j = \frac{2(1-2^{j+1})4^j B_{j+1}}{j+1}.$$

This is zero for positive even j, and its odd coefficients match $j[x^j]\log G(x)$ exactly. Hence the proposed exponential formula holds. Formal logarithmic differentiation gives the coefficient recurrence. Finally $B(z) = 2e^{-4z}/(1+e^{-4z})$ is the defining Euler-polynomial generating function at $u = 1$, with argument $-4z$. □

2 The three reciprocal and substitution identities

Theorem 2. *For $H(x) = xG(x)$ the following three conjectures hold as formal Laurent-series identities:*

$$H(x) = -\frac{x^2}{H(-x)}, \qquad H(x) = -H\left(\frac{-x}{1+4x}\right), \qquad H(x) = \frac{x^2}{H(x/(1-4x))}.$$

Proof. The logarithm derived above contains only odd powers. Thus $G(x)G(-x) = 1$, or $H(x)H(-x) = -x^2$, proving the first identity. This is a formal conclusion about coefficients; it does not evaluate the analytic gamma expression at negative t.

The gamma duplication formula gives, for positive t,

$$F(t) = t\left(\frac{\Gamma(t+1/2)}{\Gamma(t+1)}\right)^2.$$

The recurrence $\Gamma(t+1) = t\Gamma(t)$ therefore yields the exact identity

$$F(t)F(t-1/2) = \frac{t-1/2}{t}.$$

Asymptotic expansions at positive infinity are unique and may be multiplied and shifted by a fixed constant. With $x = 1/(8t)$ this becomes the formal identity

$$G(x)G\left(\frac{x}{1-4x}\right) = 1-4x.$$

Multiplying by $x^2/(1-4x)$ proves $H(x)H(x/(1-4x)) = x^2$, the third assertion. Combine it with the first to obtain $H(x/(1-4x)) = -H(-x)$. Replacing x by $-x$ proves the second assertion. All substituted series have zero constant term, so their formal compositions are defined. □

Verification and scope. The standard-library verifier computes the Euler coefficients and the proposed exponential recurrence exactly with rational arithmetic, and compares all 301 published coefficients. Independently, the exact gamma recurrence $F(t+1)/F(t) = (t+1/2)^2/(t(t+1))$ determines each next asymptotic coefficient by coefficient comparison; the verifier reconstructs all 301 terms that way as a separate control. It verifies all three identities through degree 100 using exact formal operations. The proof establishes the identities at every order. We do not claim convergence, new authorship of Stirling's formula, or OEIS acceptance of an update.

References

[1] M. Somos, OEIS A092635, including the proposals of P. Bala (2015) and P. D. Hanna (2024), https://oeis.org/A092635, accessed October 26, 2026.

[2] NIST Digital Library of Mathematical Functions, equation 5.11.1, logarithmic Stirling expansion, https://dlmf.nist.gov/5.11.E1.